Fiscal Year:	FY 2008	**Task Last Updated:**	FY 04/16/2008
PI Name:	Jacobs, Jeffrey Ph.D.		
Project Title:	An Experimental Investigation of Incompressible Richtmyer-Meshkov Instability		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/18/2002	**End Date:**	11/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	4
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Niederhaus, Charles (NASA Glenn Research Center)		
Grant/Contract No.:	NAG3-2725		
Performance Goal No.:			
Performance Goal Text:			
	Richtmyer-Meshkov instability occurs when a planar interface separating two fluids of different densities is impulsively accelerated in the direction of its normal. The focus of this research project is to observe and quantify the nonlinear development of this instability from a simple well characterized initial state into a turbulent flow, and to determine whether a self-preserving final turbulent state is achieved. The experiments utilize a novel experimental apparatus which circumvents many of the experimental difficulties that have previously limited the study of Richtmyer-Meshkov instability. In this system, the instability is generated by impulsively accelerating a rectangular box containing two liquids. The effects of gravity are minimized by keeping the fluids in free-fall. Previous NASA funding supported the construction of a laboratory apparatus in which the impulsive acceleration is imparted by bouncing the fluid box off of a fixed spring. The current project will extend the previous study and focus on the turbulent regime to determine self-similarity and to obtain the turbulent growth rate. The extended duration of microgravity offered by the 2.2 Second		

Task Description:	Drop Tower at GRC and the KC-135 Reduced Gravity Aircraft will be utilized to study the turbulent regime. An adaptation of the existing laboratory apparatus for use in the 2.2 Second Drop Tower was initiated under previous NASA funding. Experiments will therefore initially be carried out in this facility using the new apparatus and later progress to the Reduced Gravity Aircraft. The adaptation of the experiments to the KC-135 will be aided by experiments at the University of Arizona that will focus on developing techniques for initial interface formation using paramagnetic fluids. These methods will also prove useful for a possible future space experiment. The objective of the funded research is to carry out ground-based experiments to study the nonlinear development and transition to turbulence of Richtmyer-Meshkov instability with the goal of quantifying the development of the resulting turbulent flow and to determine if a self-similar state is achieved. The long term objective of this project is to develop methods and procedures for a possible future flight experiment in order utilize the low gravity environment of space for the creation of an experiment with virtually unlimited duration to study the fully turbulent Richtmyer-Meshkov instability. Richtmyer-Meshkov instability is one of the most fundamental of fluid instabilities and is of importance in fields ranging from astrophysics to materials processing. For example, Richtmyer-Meshkov instability is of critical importance to inertial confinement fusion. The degree of compression capable in laser fusion experiments is ultimately limited by Richtmyer-Meshkov instability and another similar instability called Rayleigh-Taylor instability. Thus, these instabilities represent the single most significant barrier to attaining positive-net-yield fusion reactions in laser fusion facilities. Richtmyer-Meshkov instability is also of importance to high speed combustion applications, where, for example, it can greatly enhance mixing and thus significantly alter the burning rate. Knowledge gained from this study should aid in the design of laser fusion targets and supersonic combustors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Richtmyer-Meshkov instability is one of the most fundamental of fluid instabilities and is of importance in fields ranging from astrophysics to materials processing. For example, Richtmyer-Meshkov instability is of critical importance to inertial confinement fusion. The degree of compression capable in laser fusion experiments is ultimately limited by Richtmyer-Meshkov instability and another similar instability called Rayleigh-Taylor instability. Thus, these instabilities represent the single most significant barrier to attaining positive-net-yield fusion reactions in laser fusion facilities.
Task Progress:	The funds from this project were used to support three experimental programs: (1) Experiments on the single-mode incompressible Richtmyer-Meshkov (RM) instability in the University of Arizona Drop Tower; (2) Long duration incompressible RM experiments in the NASA Glenn 2.2 Second Drop Tower; and (3) The development of new techniques for producing stable interfaces in microgravity. As part of this last task experiments were carried out on the Rayleigh-Taylor instability utilizing a magnetic field to initially stabilize a heavy magnetic fluid on top of a lighter non-magnetic one. The three-dimensional RM experiments were initiated with a single, square-mode initial perturbation. Images obtained from these experiments revealed the development of a vorticity pattern in the form of an array of vortex rings, alternatively traveling upward and downward. This observation is in contrast to the pattern observed in the two-dimensional flow which indicates a stationary row of line vortices. Amplitude measurements taken from these experiments reveal a late-time dependence similar to that of the earlier two-dimensional experiments of Niederhaus and Jacobs. In addition, the dimensionless scaling found to collapse the 2D experiments in the earlier study appears to effectively collapse the 3D results indicating that the effects of viscosity and the initial amplitude are unimportant, at least when only considering the amplitude of the instability. The 3D amplitude plots are found to differ from the 2D results in that they show significantly greater nonlinear growth, a fact in agreement with most nonlinear models of the flow. This greater growth rate also yields agreement with linear stability theory to later dimensionless time than has been observed in two-dimensional experiments. Experimental growth rates were calculated and compared with the models of Oron et al. and Goncharov. The results of these models were found to provide very good agreement with the two-dimensional experiments but significantly worse agreement with the three-dimensional measurements. The models appears to capture approximately the correct magnitude of the growth rate. However, they possesses the incorrect time dependence which can be simply explained by the differences in the vorticity distributions of the 2D and 3D flows. Comparison is also made to global amplitude models Zhang & Sohn and Mikaelian and a modified version of the model of Sadot et al. As was similarly found in comparisons with the two-dimensional measurements of Niederhaus and Jacobs, the Padé approximation of Zhang & Sohn shows poor agreement with the data at late time while the Mikaelian and modified Sadot models show better agreement. The modified Sadot model has the best agreement of the three, but only by a small amount. The long duration RM experiments were conducted utilizing the 2.2 Second Drop Tower at the NASA Glenn Research Center utilizing a drop rig designed to mimic the acceleration profile received by the fluids in the UA apparatus. The earliest experiments showed that the time duration and vibration incurred in raising the rig to the top of the tower produced unwanted mixing of the two fluids. As a result, modifications were made to the rig to allow for withdrawal of the thin mixed region at the interface just prior to the rig drop. Experiments were conducted using this configuration over a range of Reynolds numbers lower than was possible with the University of Arizona apparatus. Despite the significant difference in Reynolds number between the two sets of data the comparison is favorable. It was originally planned to perform larger Reynolds number experiments which would provide data at a much later dimensionless time. However these large Reynolds number experiments proved difficult to conduct due to safety considerations in the drop tower, and unexpected difficulties in obtaining a well-defined initial perturbation when oscillating the tank at higher amplitudes. The difficulties arose because the rig is held by a single cable, and internal oscillations would induce secondary oscillations of the rig, including undesirable out-of-plane motions. These problems also made it difficult to conduct experiments with shorter wavelengths. An important component of the proposed research focused on the development of a method to produce stable interfaces in microgravity using paramagnetic liquids and the Kelvin force arising in a magnetic field gradient. This effort began by evaluating electromagnet configurations for this purpose considering both monopole and dipole configurations. In this process different configurations were modeled using the finite element package ANSYS. Using the results of this analysis it was concluded that a dipole electromagnet configuration would best suit our purpose and a large (10” pole diameter) electromagnet with power supply was purchased. Subsequently, ANSYS was used to determine the optimum

	pole face geometry to carry out the dipole experiments. Follow on experiments utilizing this configuration demonstrated its ability to stabilize a heaver magnetic fluid on top of a lighter magnetic one, and confirmed our calculations predicting that this choice of pole geometry does generate a flat interface. Following the successful demonstration of magnetic levitation, an experimental investigation of the Rayleigh-Taylor instability utilizing this system was begun. In these experiments the liquids, a miscible combination of paramagnetic salt solution and one of three nonmagnetic solutions, were contained in a tank placed between the poles of the electromagnet. Rayleigh-Taylor instability was then initiated by rapidly removing power to the electromagnet which results in the heavy fluid falling under the influence of gravity. The resulting instability was visualized using backlit photography. Experiments initiated with an initially flat interface develop into a random surface pattern the scale of which is well approximated by the fastest growing wavelength given by viscous linear stability theory. Measurements of the mean mixing zone width were obtained from ensembles of backlit experiments which revealed growth rate measurements within the range measured in previous Rayleigh-Taylor experiments. Following the success of the backlit experiments effort was made to use PLIF to visualize the flow. To accomplish this Rhodamine 6G dye is mixed with the lower fluid and an Argon Ion laser is used to form a thin light sheet which is passed through the top of the tank, illuminating a thin cross section of the two fluid mixture. Grwoth rate measurements taken from ensemble averages of PLIF experiments revealed growth constants in excellent agreement with the backlit experiments.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Abstracts for Journals and Proceedings	Gohardani O, Oemke R, Jacobs J. "Experimental study of Rayleigh-Taylor instability utilizing a paramagnetic liquid combination." Presented at the 60th Meeting of the APS/DFD, Salt Lake City, Utah, November 18 - 20, 2007. 60th Meeting of the APS/DFD, Salt Lake City, Utah, November 18 - 20, 2007. , Nov-2007
Dissertations and Theses	Oemke R. "An Experimental Study of Rayleigh-Taylor Instability Utilizing Planar Laser Induced Fluorescence with Paramagnetic Fluids." Thesis, University of Arizona, May 2007. , May-2007